

The Impact of ChatGPT on Open-Source AI Adoption: Evidence from Hugging Face

Group 4

Drake Niepoetter

Hung-En Chen

Luisa Garate

Victor Minh Phan

Zhanhui Lin

BANA 290: Artificial Intelligence: The Impact on Firms, Society, and the Economy

May 27, 2026

1. Executive Summary

This study examines whether the public release of ChatGPT in November 2022 accelerated the adoption of open-source large language model-style text-generation models on Hugging Face, relative to non-text model categories (image classification, object detection, audio classification, and speech recognition). We compile metadata for 360,487 unique models created between January 2021 and December 2025 and apply a difference-in-differences design comparing new model creation rates for treatment (text-generation) versus control categories before and after the release. We find a large, positive, and statistically significant treatment effect. After ChatGPT's release, the rate of new text-generation model creation grew about 3.8 times faster than the rate of new non-text model creation (DiD coefficient = 1.34, $p = 0.006$). A formal parallel-trends test ($p = 0.847$) and an event-study specification both support the interpretation that this difference reflects the ChatGPT event rather than pre-existing divergence. The effect appeared roughly twelve months after the release and has continued to grow through 2024 and 2025. Overall, the results indicate a sustained reallocation of open-source AI development toward generative text models, with downstream implications for (i) economic activity and the cost of building specialized AI systems, (ii) organizational capability needs and build-versus-buy decisions, and (iii) societal concerns around equity, content authenticity, and regulatory fit. We conclude with targeted recommendations for business leaders, researchers, and policymakers.

2. Introduction

The rapid evolution of generative artificial intelligence, particularly large language models (LLMs), has dominated technology discussions over the past three years. This paper examines whether the public release of ChatGPT in November 2022 accelerated the open-source adoption of text-generation models. We focus on open-source activity because it provides a uniquely observable, time-stamped record of AI development that is difficult to obtain from commercial systems. In particular, Hugging Face hosts a large and diverse repository of publicly released models across text, image, and audio tasks; the monthly creation of new models therefore offers a high-frequency indicator of where developer attention and ecosystem investment are shifting.

Our research question is: Did the release of ChatGPT differentially increase the adoption of LLM-based text-generation models compared with non-text model categories on Hugging Face?

To answer this, we compile metadata for 360,487 unique models created between January 2021 and December 2025 and estimate a difference-in-differences design comparing changes in new model creation for text-generation (treatment) versus non-text categories (control) before and after November 2022. We find a large and statistically significant post-release shift toward text generation: after ChatGPT’s release, the creation rate of text-generation models grew at about 3.8 times the rate of non-text models (DiD coefficient = 1.34, $p = 0.006$). The remainder of the manuscript proceeds as follows: Section 3 reviews related literature; Sections 4-5 describe the data, identification strategy, and results; Sections 6–8 discuss implications; Sections 9-10 provide recommendations and concluding remarks.

3. Background and Literature Review

3.1 Large Language Models and the Foundation-Model Paradigm

Large language models (LLMs) are AI systems trained on large amounts of text to generate natural language. Modern LLMs became possible after the introduction of the Transformer architecture in 2017, which allowed models to process large-scale text data more efficiently (Wolf et al., 2020). In the early 2020s, AI development shifted from task-specific models to large foundation models that could be adapted through fine-tuning or prompt engineering (Bommasani et al., 2021). ChatGPT was the first widely adopted public product based on this approach, and this paper examines its downstream effects.

3.2 Hugging Face as the Open-Source AI Platform

Hugging Face is the leading online platform for sharing open-source artificial intelligence models. Founded in 2016 and originally focused on natural language processing, it has grown into the central repository for the open-source AI ecosystem, hosting hundreds of thousands of models, datasets, and applications across text, image, audio, and multimodal tasks (Hugging Face, 2023). The platform plays a role in AI similar to GitHub's in software: it provides distribution, version control, model cards documenting capabilities and limitations, and a social layer that helps researchers and developers discover, evaluate, and build on each other’s work. Because it aggregates such a large share of publicly released AI models, Hugging Face offers an unusually comprehensive view of open-source AI activity over time. This makes it a natural setting for empirical research on how the AI ecosystem responds to external shocks, including the release of products like ChatGPT and broader debates about open-source AI development.

3.3 Relevant Economic Theories

Two strands of economic theory help explain our results. The first is the General-Purpose Technology (GPT) framework developed by Bresnahan and Trajtenberg (1995), which describes technologies that spread across industries and support waves of innovation. This framework helps explain the rapid growth of text-generation models in our data. The second is Schumpeter's (1942) idea of creative destruction, which explains how new technologies shift resources away from older applications. This helps explain why non-text AI models continued to grow, but at a slower pace than generative AI models. Research by Acemoglu and Restrepo (2018) and Brynjolfsson, Li, and Raymond (2023) also suggest that productivity gains from AI depend on how firms redesign workflows around the technology. Our study contributes to this literature by showing large-scale growth in open-source generative AI development after the release of ChatGPT.

4. Methodology

4.1 Data Sources

We retrieved model-level metadata from the Hugging Face Hub, the largest open-source platform for sharing artificial intelligence models, using the official `huggingface_hub` Python client. The Hugging Face API provides an extensive set of attributes for each model, including the model identifier, author, creation timestamp, last modification date, download count, like count, pipeline task tag, library name, and a flexible list of descriptive tags. Our queries targeted the `pipeline_tag`, the official task label assigned by model maintainers, which specifies whether a model is designed for text generation or for non-text tasks such as image or audio processing. The final data collection, completed on May 6, 2026, yielded 429,399 unique model records from the platform.

4.2 Treatment and Control Groups

The treatment group consists of large language model-style generative text models, defined as those with a `pipeline_tag` of either `text-generation` or `text2text-generation`. The control group consists of non-text artificial intelligence task types, specifically models tagged as `image-classification`, `object-detection`, `automatic-speech-recognition`, `audio-classification`, `image-segmentation`, or `image-to-text`. These categories were chosen because they are major, long-standing tasks on the platform whose growth trends are unlikely to be directly affected by the

public release of a generative text product. Models without a recognized pipeline_tag in either group are excluded from the analysis. From the initial 429,399 records, we removed 991 observations containing a Hugging Face placeholder timestamp (dated March 2, 2022) and restricted the sample to the January 2021 through December 2025 analysis window, resulting in a final study sample of 360,487 models. Of these, 298,733 belong to the treatment group, and 61,754 belong to the control group.

4.3 Analytical Approach

We use a difference-in-differences (DiD) design to estimate the effect of ChatGPT’s November 2022 release on the adoption of open-source text-generation models. After keeping only models with valid creation dates and removing a small set of Hugging Face placeholder timestamps (records dated March 2, 2022), we group the data at the (group, calendar-month) level. The outcome variable is the log of the number of new models created in each cell. We estimate three specifications: (i) a baseline DiD with a treatment indicator, a post-November-2022 indicator, and their interaction; (ii) the same specification with month fixed effects to account for shared changes over time; and (iii) an event-study specification that interacts treatment with month indicators to show how the effect evolves over time. The main coefficient of interest is the treatment-by-post interaction, which measures the change in new-model creation in the treatment group relative to the control group after the event. A positive and statistically significant coefficient would indicate that ChatGPT’s release was followed by a disproportionate increase in the creation of text-generation models. Standard errors are heteroskedasticity-robust (HC3).

4.4 Robustness Checks

The validity of any difference-in-differences estimate depends on the parallel-trends assumption: in the absence of treatment, the treatment and control groups should follow similar paths over time. We assess this directly by regressing log monthly new-model counts on a treatment indicator, a linear time trend, and their interaction, restricting the sample to the pre-November-2022 period. A statistically insignificant interaction term supports the parallel-trends assumption. We also examine the event-study coefficients in the pre-period; coefficients close to zero further indicate that the two groups were comparable before the event.

4.5 Limitations

Several limitations should be noted. First, the study includes only eight pre-event months, limiting the statistical power of the parallel-trends test. Second, some legacy Hugging Face models contain placeholder timestamps that were excluded from the regression but still appear in descriptive figures. Third, because Hugging Face download counts reflect a rolling 30-day window rather than lifetime usage, our analysis focuses on model-creation rates instead of downloads. Moreover, many post-2023 text-generation models are fine-tuned or compressed variants of a smaller set of base models, meaning our counts include both original models and their derivatives. Finally, while the study uses non-text-generation categories as a control group, future research could incorporate additional category-specific controls such as audio and video models, which could help better account for broader AI-industry-wide growth trends that occur independently of the release of ChatGPT.

5. Results & Key Findings

This section presents the empirical results of our difference-in-differences analysis. We begin by describing the composition of our study sample and then formally test the parallel-trends assumption that underlies the design. Next, we present our main difference-in-differences estimate, which is the headline result of our analysis, followed by an event-study specification that shows how the effect changes over time. We conclude with a discussion of the size of the estimated effect and its real-world implications.

5.1 Sample Composition

Table 1 reports the number of new models created each year, broken out by treatment and control group. The treatment group, consisting of large language model-style text-generation models, grew from 8,146 new models in 2022 to 142,848 in 2025, a seventeen-fold increase over the analysis window. The control group, by contrast, grew from 7,421 to 17,610 over the same period, a roughly two-fold increase. The gap between the two groups' trajectories becomes clear starting in 2023 and widens significantly in 2024 and 2025. This descriptive pattern sets the stage for the more formal difference-in-differences analysis that follows.

Table 1. New Hugging Face model releases per year, by group.

Year	Treatment	Control	Total
2022	8,146	7,421	15,567

2023	32,362	16,441	48,803
2024	115,377	20,282	135,659
2025	142,848	17,610	160,458
Total	298,733	61,754	360,487

5.2 Parallel-Trends Test

Before turning to the main estimate, we first check whether the treatment and control groups followed similar trajectories before November 2022. Restricting the sample to the pre-event window, we regress the log of monthly new-model counts on a treatment indicator, a linear time trend, and their interaction. The interaction coefficient is small and not statistically significant ($\beta = -0.054$, $p = 0.847$), meaning we cannot reject the null hypothesis that the two groups were on parallel paths before the event. We interpret this as supporting the parallel-trends assumption required for a valid difference-in-differences interpretation. At the same time, the pre-event window contains only eight months, which limits the statistical power of this test. For this reason, we also rely on the event-study specification in Section 5.4, where pre-event coefficients close to zero provide additional evidence that the two groups were comparable before the event.

5.3 Main Difference-in-Differences Estimate

Table 2 reports the results of our baseline difference-in-differences specification, using the log of monthly new-model counts as the outcome. The treatment-by-post interaction is the coefficient of interest. The estimate is 1.343 with a heteroskedasticity-robust standard error of 0.486, corresponding to a p-value of 0.006 and a 95 percent confidence interval that excludes zero. This indicates that, after ChatGPT's release in November 2022, the rate of new text-generation model creation increased significantly faster than the rate of new model creation in the control group. The pre-treatment difference between the groups, captured by the treatment coefficient alone, is small and statistically indistinguishable from zero ($\beta = 0.015$, $p = 0.974$), which is consistent with the parallel-trends evidence reported above.

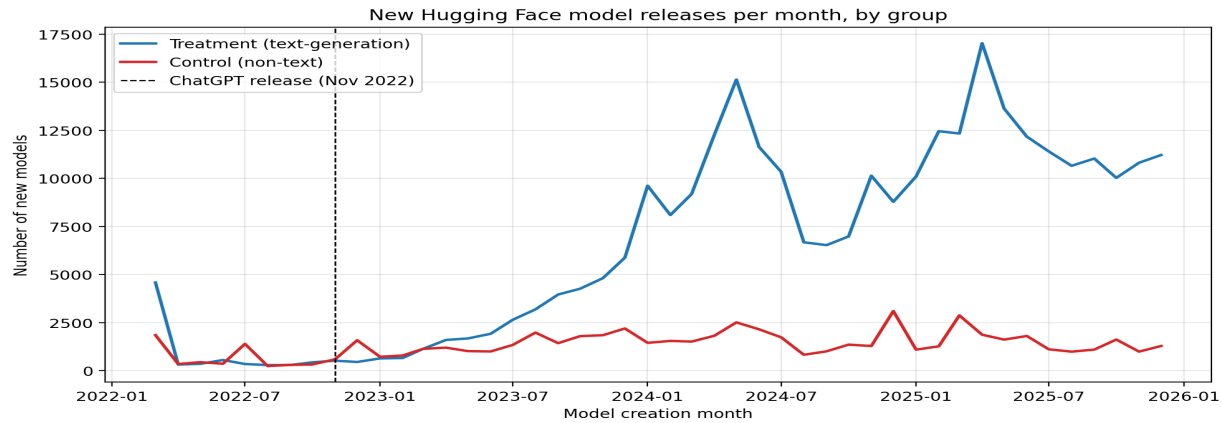
Table 2. Baseline difference-in-differences estimates. The dependent variable is the log of new models created per month.

Variable	Coefficient	Std. Error	p-value
----------	-------------	------------	---------

Intercept	6.192	0.286	<0.001
Treatment	0.015	0.451	0.974
Post-ChatGPT	1.045	0.292	<0.001
Treatment x Post (DiD)	1.343	0.486	0.006

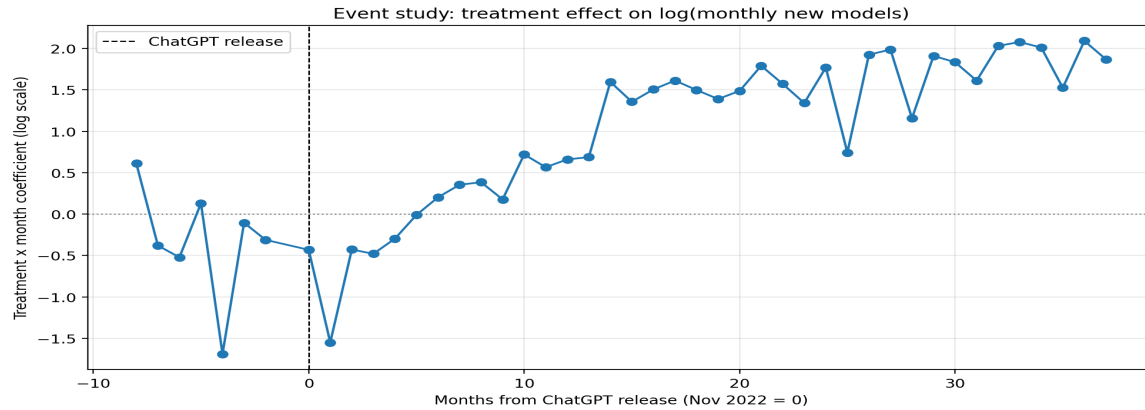
Note: Heteroskedasticity-robust (HC3) standard errors in column 3. The Treatment x Post interaction is the difference-in-differences estimate.

Figure 1, which plots monthly new-model creation by group, shows the same pattern visually. Before November 2022, the two series moved together at a similar rate. After November 2022, the treatment series begins to rise while the control series remains essentially flat. The vertical gap between the two lines widens steadily through 2024 and 2025, peaking at more than 15,000 new text-generation models in a single month, while the control group remains below 3,000.



5.4 Event Study: Dynamic Effects

The baseline difference-in-differences model estimates an average post-treatment effect but does not show how the effect changes over time. To examine these dynamics, we estimate an event-study model with separate monthly coefficients relative to November 2022. Three patterns appear in Figure 2. First, the pre-event coefficients remain close to zero, supporting the parallel-trends assumption. Second, the effect does not appear immediately and takes about twelve months to grow meaningfully, likely reflecting the time needed to develop and release new models. Third, the effect continues increasing through 2024 and 2025 rather than fading. By the end of the sample period, the treatment effect exceeds 2.0 on the log scale, suggesting substantially higher treatment-group activity after the release of ChatGPT.



5.5 Magnitude and Interpretation

Because the outcome variable uses log count, the difference-in-differences coefficient represents a proportional change. Converting the estimate out of logs gives: $e^{1.343} \approx 3.83$. This means that after the release of ChatGPT, the creation of new text-generation models grew at about 3.8 times the rate of the non-text control group. The treatment group increased from about 8,100 models in 2022 to nearly 143,000 in 2025, while the control group grew from about 7,400 to 17,600. The results provide strong evidence that a large and sustained shift toward open-source generative text models followed the release of ChatGPT. This pattern appears in the descriptive statistics, the difference-in-differences model, and the event-study results.

6. Economic Impact Analysis

Our findings suggest a substantial shift in capital, attention, and developer effort toward generative artificial intelligence. Three economic mechanisms help explain how a single product launch could produce such a large and sustained change in open-source activity, and what those mechanisms imply for market dynamics, cost structures, and industry disruption.

6.1 ChatGPT as a General-Purpose Technology Shock

Economists Bresnahan and Trajtenberg (1995) define a General-Purpose Technology (GPT) as a technology that is widely useful across sectors, improves quickly, and supports complementary innovation. Although ChatGPT is not hosted on Hugging Face, its release reduced the difficulty of experimenting with large language models and signaled that generative AI could become a foundation for downstream applications. The 142,848 text-generation models created in 2025 are not 142,848 competing products, but rather complementary innovations, including fine-

tunes, smaller, more efficient versions, and specialized variants built on a small set of base models.

6.2 Falling Marginal Cost of AI Product Development

Before the foundation-model era, building an artificial intelligence system required large datasets, significant computing power, and specialized machine-learning expertise. Open-source large language models on Hugging Face have reduced these costs by enabling developers to fine-tune pretrained models rather than train systems from scratch. Developers can now create specialized AI applications in hours rather than months using smaller datasets and lower-cost hardware. The rapid growth of fine-tunes and lightweight text-generation models after 2023 suggests that the marginal cost of AI development has fallen substantially. Economically, this lowered barriers to entry and expanded participation to smaller firms, researchers, and independent developers that previously lacked the resources to compete in the AI industry.

6.3 Capital Reallocation and Industry Disruption

The asymmetry between the treatment and control groups in our analysis reflects a broader reallocation of capital across the artificial intelligence industry. Venture funding for generative-AI startups rose nearly eightfold from 2022, reaching \$25.2 billion in 2023 (Stanford HAI, 2024), while funding for non-generative AI categories grew more slowly. This pattern aligns with Schumpeter's (1942) concept of creative destruction, in which new technologies redirect investment and innovation away from older paradigms toward emerging opportunities. Although non-text AI categories such as image classification and object detection continue to grow, generative AI has increasingly become the primary focus of developer attention, research funding, and ecosystem investment.

7. Organizational Impact Analysis

The relative growth in text-generation model creation reflects more than an industry-level reallocation of capital; it also signals a substantial reorganization of work inside firms that depend on artificial intelligence. As large language models have become more accessible, firms now face new decisions about workforce skills, AI adoption strategies, and infrastructure development. These organizational changes are shaping how companies compete and adapt in an increasingly AI-driven economy.

7.1 New Skills and Emerging Roles

The rise of generative AI has created demand for new skill sets, including prompt engineering, model fine-tuning, and AI output evaluation. A 2023 study of generative AI in customer-support work found that access to an AI assistant raised the productivity of less-experienced workers by 34 percent, improving overall service quality (Brynjolfsson, Li, & Raymond, 2023). These studies suggest that generative AI does not simply displace workers; it shifts the composition of valuable skills, shifting which workers benefit most from new tools. In response, employees are being trained and teams are reorganized around AI capabilities rather than traditional functional structures.

7.2 The Build-Versus-Buy Decision and Implementation Challenges

Our data show that a large share of post-2023 text-generation models on Hugging Face are derivatives of a few open base models, reflecting a broader shift towards fine-tuning open-source models rather than building a proprietary model from scratch or paying for commercial application programming interfaces (APIs). This strategy allows companies to maintain greater control over their data and reduce long-term costs, but it also requires in-house machine-learning expertise, infrastructure and supporting systems such as data pipelines, monitoring tools, guardrails, and evaluation procedures. This also creates challenges for firms deciding when to update or migrate systems. As a result, the productivity benefits of generative AI depend not only on access to models, but also on investments in workflows, employee training, and organizational adaptation, giving firms that integrate these capabilities effectively a stronger long-term advantage.

8. Societal Consequences

The patterns documented in our data raise questions that extend beyond markets and firms to society as a whole: more of everyday digital infrastructure, including search, writing tools, customer service, education technology, and creative production will be powered by large language models in the coming years. There are three societal consequences of that shift: concerns about equitable access and representation, risks related to misinformation and content authenticity, and emerging questions for policy and regulation.

8.1 Access, Equity, and Representation

Since open-source distribution lowers the cost of access to powerful AI tools, there is a clear equity gain for developers in regions without major technology firms. However, the underlying

base models are produced by a small number of well-resourced organizations, overwhelmingly trained on English-language and Western-context data. Bommasani et al. (2021) note that this creates downstream risks of homogenized outputs and propagated biases. The open-source surge expands who can build with AI, but does not necessarily expand whose languages, communities, and contexts are well represented within it.

8.2 Misinformation and Content Authenticity

A natural consequence of cheap, widely available text generation is a sharp increase in synthetic content online. The same fine-tuning workflows that let a small firm build a helpful customer-service assistant also allow bad actors to create persuasive misinformation, fake product reviews, or impersonation content at scale. The fastest growth in our treatment group comes from fine-tunes and lightweight variants which can be adapted for specific tasks, including harmful ones, with relatively little technical skill. Detection tools, watermarking methods, and platform-level moderation have not adapted to new generative models, creating a growing gap between how easy it is to produce synthetic content and how difficult it is to identify it, raising debates about misinformation and content authenticity.

8.3 Policy and Regulation

The rapid growth of open-source generative AI creates challenges for regulation. Open-source AI makes accountability more difficult than in closed systems because models can be modified by many users. Policymakers now face a difficult balance between protecting innovation and reducing social risks. Clear limitations and accountability mechanisms are needed for how AI capabilities are deployed in business settings. Its long-term social impact will depend on how generative AI is regulated and used in the future.

9. Strategic Recommendations

Our findings point to several practical takeaways for three groups of stakeholders. The recommendations below draw directly from the patterns in our data and from the broader arguments developed in Sections 6 through 8.

For Business Leaders

Firms should view the open-source large language model ecosystem as a meaningful source of competitive advantage, not a side project. The data show that fine-tuning an open base model is now a practical alternative to building a proprietary system from the ground up. To use these

models effectively, leaders need to invest in the infrastructure that supports safe deployment, including evaluation pipelines, monitoring tools, and staff training. These supporting investments often determine whether productivity gains actually materialize, putting them in a stronger position to adopt new capabilities as base models continue to improve.

For Researchers

The pattern we document opens several useful directions for future research. Scholars should examine how open-source generative AI affects productivity, how reliance on a small number of base models influences downstream innovation, and how AI regulations designed for closed systems can adapt to more distributed forms of development. Longer time horizons will also be important as the post-event window expands, since the effect could continue to fluctuate over time. These questions point toward a broader agenda for studying open-source AI development and its long-run economic and social effects.

For Policymakers

Policymakers should recognize that open-source generative AI spreads in ways that traditional governance tools were not designed to handle. Rules written only for closed commercial systems will miss a large and rapidly growing share of real activity. Effective frameworks should focus on the points where intervention has the most impact, including the small number of upstream base models, content-authentication standards that can scale with rising volumes of synthetic content, and clear rules for assigning responsibility for downstream harms. Policy will also need to balance safety with the risk of pushing generative AI development out of open ecosystems and into the hands of a few large firms, thereby reducing competition and limiting equitable access.

10. Conclusion

This study examined whether the release of ChatGPT accelerated the growth of open-source text-generation models on Hugging Face compared with non-text models. Using a difference-in-differences analysis on 360,487 models from 2021 to 2025, we found a large and statistically significant increase. After ChatGPT's release, text-generation model creation grew about 3.8 times faster than the control group. The pre-trends and event-study tests also suggest that this change was linked to the ChatGPT event rather than existing trends. The results suggest a major shift toward open-source generative AI development, which may affect investment patterns, workplace organization, and discussions about fairness and governance. The study is limited to a

single platform and counts fine-tuned models separately, but the main findings remain consistent across tests. The rapid and lasting growth of open-source AI after ChatGPT's release shows how strongly a single technological event can influence the industry. The long-term impact will depend on how companies, researchers, and policymakers respond in the future.

References

- Acemoglu, D., & Restrepo, P. (2018). *Artificial intelligence, automation, and work* (NBER Working Paper No. 24196). National Bureau of Economic Research.
<https://www.nber.org/papers/w24196>
- Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., ... Liang, P. (2021). *On the opportunities and risks of foundation models*. Stanford Center for Research on Foundation Models. <https://crfm.stanford.edu/assets/report.pdf>
- Bresnahan, T. F., & Trajtenberg, M. (1995). General purpose technologies; ‘Engines of growth’? *Journal of Econometrics*, 65(1), 83-108. [https://doi.org/10.1016/0304-4076\(94\)01598-T](https://doi.org/10.1016/0304-4076(94)01598-T)
- Brynjolfsson, E., Li, D., & Raymond, L. (2023). *Generative AI at work* (NBER Working Paper No. 31161). National Bureau of Economic Research.
<https://www.nber.org/papers/w31161>
- Hugging Face. (2023). *Hugging Face model hub*. <https://huggingface.co/models>
- Schumpeter, J. A. (1942). *Capitalism, socialism and democracy*. Harper & Brothers.
- Stanford Institute for Human-Centered Artificial Intelligence. (2024). *Economy: The 2024 AI Index report*. Stanford University. <https://hai.stanford.edu/ai-index/2024-ai-index-report/economy>
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Le Scao, T., Gugger, S., ... Rush, A. M. (2020). Transformers: State-of-the-art natural language processing. *Proceedings of the 2020 Conference on Empirical Methods*

in Natural Language Processing: System Demonstrations, 38-45.

<https://aclanthology.org/2020.emnlp-demos.6>